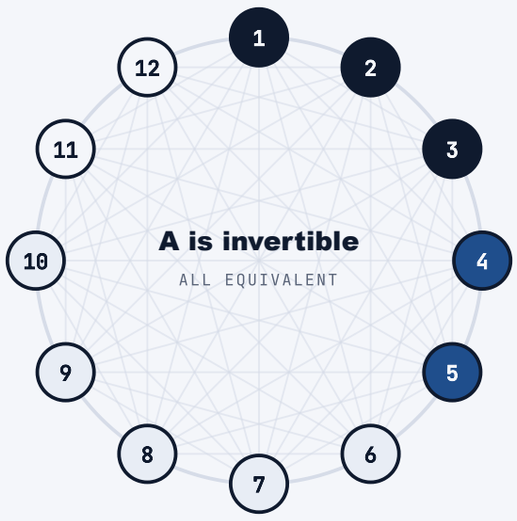
SQUARE MATRICES · TFAE

The invertibility equivalence

Twelve statements about an $n \times n$ matrix, no one of them primary. Any may be taken as the definition and the rest derived — which is why the picture is a ring and not a list.

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STATEMENTS



1	$\det(A) \neq 0$	DET
2	A is a product of elementary matrices	DET
3	0 is not an eigenvalue of A	DET
4	$\operatorname{rank}(A) = n$ — full rank	RANK
5	$\operatorname{rref}(A) = I_n$	RANK
6	The columns of A are linearly independent	SPACE
7	The columns of A span $\mathbb{R}^n$	SPACE
8	The columns of A form a basis for $\mathbb{R}^n$	SPACE
9	The rows of A are linearly independent	SPACE
10	$\operatorname{null}(A) = \{\mathbf{0}\}$	SPACE
11	$A\mathbf{x} = \mathbf{0}$ has only the trivial solution	SYSTEM
12	$A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$	SYSTEM

WHAT THE TAGGING EXPOSES

The twelve are not twelve independent claims so much as one fact seen from four sides. $\operatorname{rank}(A) = n$ and $\operatorname{rref}(A) = I_n$ are the same statement in two notations, and the row condition follows from the column condition because row rank equals column rank. What the tagging exposes is how much of the theorem lives in the subspace corner — five of twelve — against a determinant corner that gets stated first and remembered longest.

det ×3 · rank ×2 · space ×5 · system ×2